

Skills Summary

Identifying Conics and Their Key Features

Use this reference while completing your worksheet or when studying.

Skill 1 — Naming a conic from its equation

Put both squared terms on one side and call their coefficients A (for x^2) and C (for y^2). Then the product $A \cdot C$ names the curve: positive and equal coefficients means a circle, positive and unequal means an ellipse, negative means a hyperbola. If one squared term is missing altogether, it is a parabola.

Why: same signs let the two terms trade off against a fixed total, so the curve closes; opposite signs let one grow without bound, so it opens into two branches.

Example: Name the conic $3x^2 + 3y^2 = 27$.

Step 1. Both squared terms are present, so the sign test decides it — multiply the two coefficients before looking at anything else.

Step 2. $A = 3$ and $C = 3$, so $A \cdot C = 9$, which is positive, and the coefficients are equal.

$\Rightarrow A \cdot C = 9$, positive with $A = C$: a **circle** (radius 3 after dividing through).

Try it: Compute $A \cdot C$ for $5x^2 - 2y^2 = 20$ and name the conic.

check: — , negative, so a hyperbola

Skill 2 — Centre and radius of a circle

Standard form: $(x - h)^2 + (y - k)^2 = r^2$, centre (h, k) , radius r . The right-hand side is r^2 , so take a square root before reporting the radius.

From the expanded form $x^2 + y^2 + px + qy + f = 0$: halve and negate each linear coefficient to get the centre, then $r^2 = (p/2)^2 + (q/2)^2 - f$.

Example: Find the centre and radius of $x^2 + y^2 - 4x + 8y + 4 = 0$.

Step 1. The squared terms have no visible centre, so complete the square in each variable — half of each linear coefficient is what moves inside the bracket.

Step 2. Half of -4 is -2 and half of 8 is 4 : $(x - 2)^2 + (y + 4)^2 = -4 + 4 + 16$.

$\Rightarrow$ centre $(2, -4)$, radius 4

Try it: Find the centre and radius of $x^2 + y^2 - 6x + 2y - 6 = 0$.

check: centre $(3, -1)$, radius 4

Skill 3 — The vertex of a parabola

Axis of symmetry first: for $y = ax^2 + bx + c$ the vertex sits at $x = -\frac{b}{2a}$. Substitute that x back into the equation to get the y -coordinate — never guess it from the constant term.

Vertex form $y = a(x - h)^2 + k$ then reads the vertex off directly as (h, k) .

Example: Find the vertex of $y = x^2 - 4x + 7$.

Step 1. The vertex lies on the axis of symmetry, and that axis depends only on a and b , so compute it first and let the equation supply the height.

Step 2. $x = -\frac{-4}{2(1)} = 2$, then $y = 2^2 - 4(2) + 7 = 4 - 8 + 7$.

$\Rightarrow$ vertex $(2, 3)$, so $y = (x - 2)^2 + 3$

Try it: Find the vertex of $y = x^2 + 10x + 21$.

check: $(-5, -4)$

Skill 4 — Ellipse and hyperbola features

Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$: vertices at distance a along the axis with the larger denominator, and foci at c with $c^2 = a^2 - b^2$ — *subtract*, because the foci sit inside.

Hyperbola $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$: the positive term names the opening direction, vertices at distance a , asymptote slopes $\pm a/b$.

Example: For $\frac{x^2}{16} + \frac{y^2}{7} = 1$, find the semi-major axis and the focal distance.

Step 1. The larger denominator sits under x^2 , so the long axis is horizontal — that decides which letter is a before any arithmetic.

Step 2. $a = \sqrt{16}$, and $c = \sqrt{16 - 7} = \sqrt{9}$.

$\Rightarrow a = 4, \quad c = 3$ (foci inside the vertices, as required)

Try it: For $\frac{y^2}{9} - \frac{x^2}{16} = 1$, find the vertex distance and the positive asymptote slope.

check: $\frac{4}{3}$ and $\frac{3}{4}$